S. K. PRADARSHAN

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PROFESSIONAL SUMMARY

Results-oriented computer Science Engineering student with expertise in Full-Stack Development, Blockchain and AI Engineering. Proven trach record in architecting decentralized forensic protocol and real-time emergency systems.

TECHNICAL SKILLS

- **Languages:** C++, Python, JavaScript, Java.
- **Frameworks:** React, Redux, Next.js, Express.js, Tailwind CSS, Material-UI.
- **Tools:** Git, Linux, Power BI, Tensorflow.js, IPFS, zk-SNARKs, WebSocket API.

EXPERIENCE

Research & Development Software Intern (Bibliotech**)** *Sep 2025 – Nov 2025*

- **Optimized** legacy codebases for internal diagnostic tools, resulting in a significant increase in processing efficiency and system reliability.
- **Developed and validated** complex testing models for **focsem (autofocus)** and powerfocus systems, reducing hardware calibration errors.
- **Streamlined** development environments by managing **Python Package Index (PIP)** distributions and authoring technical documentation for experimental models.

Event Organizer (CodeRizz Club**)** *Aug 2024 – Present*

- Orchestrated comprehensive event logistics for 5+ technical workshops, serving audiences exceeding 200 participants, encompassing venue selection, catering, A/V setup, and onsite support while adhering to strict budget guidelines.

KEY PROJECTS

SafeCrisisRouting | Python, Fast API, Mapping API [Winner, Zyren'26**]** *Mar 2025 – Dec 2025*

- Engineered **a unified disaster response platform** to synchronize real-time situational data and emergency resources, securing 1st place in a national field of 60+ teams.
- Architected a dynamic **SafeRouting engine** that utilizes survival-centric heuristics to calculate optimized evacuation and responder paths, prioritizing hazard avoidance over traditional distance-based metrics.
- Optimized a high-concurrency backend architecture to manage **live geospatial data streams**, establishing a **mission-critical command-and-control interface** for large-scale crisis management.

VitalsNet | React Native, C++, Java, Firebase, BLE, rPPG [Winner, HERitage'26] *Feb 2026 - Present*

- Engineered an Android-first deepfake detection platform using rPPG **bio-signal extraction** via TS-CAN inference to cryptographically anchor liveness proof to captured media.
- Designed a BLE-based "**Consent Handshake**" protocol enforcing real-time, proximity-aware consent validation, eliminating unauthorized recording at the protocol layer.
- Built a **multi-signal Reality Score engine** combining heartbeat authenticity, watermark verification, and device fingerprinting via a C++ native bridge for tamper-evident media provenance.

EDUCATION & CERTIFICATIONS

- Bachelor of Engineering in Computer Science and Engineering (SRM VEC) *2023 – Present*
  CGPA: 8.66 | Relevant Coursework: Algorithms, AI Engineering, Full-Stack Web Development

- ML Specialization (Stanford), AI Engineering (IBM), AI Essentials (Google)

ACHIEVEMENTS

- 1st Place in Hackathon – Zyren'26 (SRM IST Ramapuram), HERitage (SRM IST Katankulathur)
- 1st Place in Coding event – Mind Grid (CSI SRM VEC), Tech Quest (CIT), Algo rift (Sairam Institute)
- 1st Place in Paper Presentation and mini hackathon (SRM VEC)
- Honorable Mention, Model United Nations (General Assembly)
- 2nd Runner Up Inter-College Badminton Tournament and State level chess player.